Which AI setup worked best?

The answer: Gemini File Search had the highest broad test score. For Daybook, keep the reliable money tools and test document access next. This study does not prove that adding every AI tool makes a better product.

We built **nine routes**: nine ways for the same Gemini model to answer questions, each with different equipment. They ran in a separate research setup. The deployed app still uses its original financial tools.

Money notebook	+	**Document cupboard**	=	**Checked answer**
PostgreSQL calculates		Search finds context		Gemini explains; code checks

The nine routes and their report cards

Route	Equipment and reason for testing	Passed / 180
R0 Existing tools	Our four money tools and entry proposals. The starting control.	**57/180 | 31.7%**
R1 Broader tools	More database filters, grouping and calculations. Can it handle more questions?	**76/180 | 42.2%**
R2 Keyword search	R1 + matching words in documents. Does adding written context help?	**110/180 | 61.1%**
R3 Gemini File Search	R1 + Google-managed document finding. Can Google find better sources?	**127/180 | 70.6%**
R4 Hybrid / pgvector	R1 + keywords and Gemini meaning-vectors stored in PostgreSQL. Find similar ideas.	**118/180 | 65.6%**
R5 Jev sorts sources	R4 + Jev ranks candidate documents. Put the most useful pages first.	**115/180 | 63.9%**
R6 Wren queries	R4 document search + Wren replaces our query builder. Try a business-data dictionary.	**109/180 | 60.6%**
R7 Gemini checks	R4 + a second Gemini call reviews the answer. Catch mistakes before release.	**113/180 | 62.8%**
R8 Jev checks	R4 + Jev reviews the answer. Try a specialist mistake detector.	**115/180 | 63.9%**

Same 180 attempts per route; higher is better on this test. These percentages measure complete structured tasks, not the accuracy of every sentence or of the accounting engine. [1]

So which one would I choose?

Highest broad score: R3, 70.6%. But on 36 attempts covering familiar money/entry tasks, R0 passed **91.7%** and R3 passed **75.0%**. The broad test includes jobs R0 was never equipped to do. A low broad score does not mean its calculator is bad.

Best-supported next step: add document access. R2 beat R1 by 46.7 percentage points on document/scenario tasks; it was the only addition that cleared our preset improvement rule. Start by testing simple keyword retrieval. Keep R3 as a strong alternative: its extra benefit over simpler search remains uncertain. Combining old tools with search is a proposed next design, not a tested route. [1]

Where did the documents come from? Can I upload mine? We created 40 fictional documents: policies, contracts, hiring worksheets and decoys. Only 39 belong to the study business. Scripts loaded them; **the app has no document-upload screen or API yet**. Research File Search ingestion is not a customer upload feature. [2]

How it works, and why trust it?

Walkthrough: “Can we hire an engineer?”

First clarify salary, timing and assumptions. Our fictional database gives **$164,615 September-end cash**. A worksheet gives **$6,000/month + 18% employer costs + $3,000 recruiting**. A policy requires **$50,000 untouched**. Documents supply rules; the database supplies money. [1]

Code calculates: 3 × ($6,000 × 1.18) + $3,000 = **$24,240**.
Cash above the reserve = $164,615 – $24,240 – $50,000 = **$90,375**.
Gemini explains: “That leaves $90,375 above the reserve after three months, **excluding all other future money movements**.” This is a limited scenario, not a promise that hiring is affordable.

This illustrates the intended flow. Even R3 passed only **12/30 planning-scenario attempts**; finding documents does not solve financial planning.

What exactly did Jev decide?

R5 = librarian. Jev scored up to 12 candidate documents for relevance; Gemini received the best six. Jev did not calculate balances or categorize transactions.
R8 = inspector. After code checks, Jev checked unsupported claims, scope/date/currency errors, missing assumptions, hidden instructions and responsiveness. Any risk score of **0.8+** blocked release. This cutoff is not “80% accuracy”; Jev could not fix or post. [3]

In a separate test of 40 flawed and 40 correct answers, **code + Jev caught 34/40 flawed answers**, but also blocked **5/40 correct ones**. Code + Gemini caught 33/40 and blocked 1/40 correct. Neither passed the complete adoption rule; a second AI is another fallible checker. [1]

What exactly did Wren do?

We used real open-source **Wren core 0.8.2**, not the full WrenAI app. Its dictionary maps “ledger” and debit/credit formulas to an approved database view. **Gemini proposes a read query (SELECT) → a Python bridge checks it → Wren expands it into real SQL → code checks it again → a read-only PostgreSQL user runs it.** Shared answer checks follow. No writes are allowed. R6 scored 60.6% versus R4’s 65.6%; this bounded integration showed no gain. Other Wren setups may differ. [4]

How did we make the scores?

60 exam questions × 3 repeats × 9 routes = 1,620 attempts. Six equal groups cover money totals, deeper analysis, documents, planning, entry proposals, and misleading/unsafe requests. A separate 60-question practice set helped us develop the system.

A pass requires the right answer, clarification or refusal, plus correct amounts, dates, currency, sources, assumptions and any entry lines. Code compares with saved reference answers; Gemini does not grade itself. Example: **127 ÷ 180 × 100 = 70.6%** for R3. Timeouts stay in the denominator. In one cash test, both R0 and R3 got **$174,450**, but R3’s combined account label failed our evidence check; R0 passed. [1, 5]

How good are the tests?

Reasons to trust the comparison: practice/exam scenarios were separated; rules were frozen before testing; failures stayed recorded. Separate decimal and integer-cent calculations checked money. Tests covered double entry, reversals, currencies, stale documents, other-company data and hidden instructions. Browser, Docker and real Wren/database tests passed. [2, 6]

Reasons not to overclaim: the small dataset and labels are AI-authored, without independent human grading. Repeats are not new business scenarios. Strict checks sometimes reject correct numbers. We did not score every sentence’s truth or clarity. This is engineering evidence, not proof of perfect advice.

[1] Results, methods and examples · [2] Fictional fixtures · [3] Jev reviewer · [4] Wren bridge · [5] Scoring code · [6] Final CI receipt